

1 Sparse input

Measured binaural
HRTFs / HRIRs
at Q directions

Target queries q
+ sparse observed
magnitudes and xyz

2 MCA baseline

Time alignment + spatial interpolation

Magnitude correction from
smoothed magnitude

MCA magnitude M_{MCA}
+ correction feature C_{MCA}

MCA phase + original
outside-bin complex values

3 FSC correction

Frozen FSC model
FiLM-SIREN +
spectral CNN

Normalized residual
 $\tilde{r}_H(q, f)$

Denormalize

$\hat{r}_H \text{ dB} = \sigma(r) \tilde{r}_H + \mu(r)$

4 Spectrum assembly

Correct 463 selected bins
 $M_{\hat{H}} = M_{MCA} + \hat{r}_H \text{ dB}$

Selected-bin complex spectrum
 $10^{(M_{\hat{H}}/20)} \exp(j \phi_{MCA})$

Reuse 50 unpredicted
MCA complex bins

513-bin single-sided spectrum

5 Interpolated output

Conjugate-symmetric
extension

1024-point IFFT

Retain first
256 samples

Dense binaural
HRIRs / HRTFs